Current Oncology Reports (2025) 27:95–111
https://doi.org/10.1007/s11912-024-01627-1

REVIEW

Artificial Intelligence and Cancer Health Equity: Bridging the Divide or Widening the Gap

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Accepted: 21 November 2024 / Published online: 3 January 2025
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Abstract

Purpose of Review This review aims to evaluate the impact of artificial intelligence (AI) on cancer health equity, specifically investigating whether AI is addressing or widening disparities in cancer outcomes.

Recent Findings Recent studies demonstrate significant advancements in AI, such as deep learning for cancer diagnosis and predictive analytics for personalized treatment, showing potential for improved precision in care. However, concerns persist about the performance of AI tools across diverse populations due to biased training data. Access to AI technologies also remains limited, particularly in low-income and rural settings.

Summary AI holds promise for advancing cancer care, but its current application risks exacerbating existing health disparities. To ensure AI benefits all populations, future research must prioritize inclusive datasets, integrate social determinants of health, and develop ethical frameworks. Addressing these challenges is crucial for AI to contribute positively to cancer health equity and guide future research and policy development.

Keywords Artificial intelligence · Cancer health equity · Health disparities · AI bias · Precision medicine · Predictive analytics

Introduction

Background

Cancer is a leading cause of death with significant disparities across populations driven by a complex interplay of factors, including socioeconomic status, race, ethnicity, geographic location, lifestyle, genetic, immunobiological factors, and access to healthcare services [1, 2]. Despite advances in cancer research and treatment, these inequities persist, underscoring the urgent need for innovative approaches to achieve more equitable cancer care.

In this context, artificial intelligence (AI) has emerged as a powerful tool in healthcare, with the potential to revolutionize cancer diagnosis, treatment, and management [3]. AI technologies, including machine learning, deep learning, and natural language processing, have shown promise in enhancing the accuracy of cancer detection, optimizing treatment plans, and predicting patient outcomes [4, 5]. By analyzing vast amounts of data that would be impossible for humans to process manually, AI has the potential to drive precision medicine, offering personalized care tailored to individual patient characteristics — such as genetic profiles, lifestyle factors, and environmental exposures [6, 7].

Rationale

While AI holds significant promise for improving cancer care, concerns remain that it could inadvertently exacerbate existing health disparities. AI models rely on biased datasets that underrepresent certain racial and ethnic groups, leading

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to less accurate outcomes for these populations when applied in healthcare settings [8–12]. Additionally, socioeconomic factors may limit access to AI-driven technologies, potentially widening the gap between those who benefit from AI and those who do not [11]. Therefore, it's essential to evaluate whether AI is addressing cancer health disparities or contributing to them.

Objectives

This review examines the current literature on AI in cancer care, with a specific focus on health equity. We aim to determine whether AI is bridging or widening the divide in cancer outcomes and to identify gaps in current research. The review highlights promising applications of AI promoting equity and provides recommendations for future research and policy to ensure AI becomes a tool for advancing health equity in cancer care.

AI in Cancer Diagnosis

Overview of Recent Advancements in AI for *Cancer* Diagnosis

AI has made significant strides in the field of cancer diagnosis, offering new tools that can improve the accuracy and speed of detecting various cancers. Machine learning and deep learning algorithms, particularly convolutional neural networks (CNNs) have demonstrated remarkable accuracy in detecting cancers from medical imaging, such as mammograms, CT scans, and MRIs [1–3, 10, 12]. For example, CNNs have been used to identify breast cancer in mammograms with accuracy comparable to, and sometimes exceeding, that of experienced radiologists [4, 13]. Additionally, AI has been applied to digital pathology, where it aids in the analysis of tissue samples to detect various types of cancer, including lung, prostate, and skin cancers [5, 14].

Analysis of Studies Assessing the Accuracy and Efficiency of AI Tools in Diverse Populations

Despite the promising advancements in AI for cancer diagnosis, concerns about the generalizability of these tools across diverse populations have been raised [15–17]. Additionally, the underrepresentation of minority populations in the data used to train AI models has led to tools that may perform less accurately for these groups, potentially resulting in higher misdiagnosis rates or delayed treatments [8–11, 17–19].

Many AI algorithms are trained on datasets that predominantly include data from patients of European descent, potentially leading to reduced accuracy when applied to other racial and ethnic groups. A study by Yala et al. (2021) showed that while AI risk models for breast cancer detection performed well in general, the mammography-based deep learning model they developed trained with large diverse datasets, performed better across different high-risk demographic groups [13]. The model improved upon the traditional risk to capture additional characteristics and multiple time points of risk assessments to improve accuracy. This suggests that AI tools have the potential to be equally effective for all populations, potentially exacerbating existing disparities in cancer diagnosis.

Disparities in the Development and Deployment of AI Diagnostic Tools

The disparities in AI diagnostic tools are not only a result of biased datasets but also reflect broader systemic issues in healthcare [6, 7, 15, 16]. For instance, healthcare facilities serving underrepresented or low-income populations may lack the resources to implement cutting-edge AI technologies.

Moreover, there is a lack of diversity among the developers and researchers working on AI, which may contribute to blind spots in the development process [11, 20]. Addressing these disparities requires a concerted efforts, which could entail the use of a mix anonymized data sharing agreements and synthetic data, to create more inclusive datasets, engage diverse stakeholders in AI development, and ensure that AI tools are accessible and contextually appropriate to all healthcare settings, regardless of the population served.

AI in Treatment Decision-Making, Drug Discovery, and Personalized Cancer Therapies

AI has the potential to revolutionize treatment planning and personalized medicine in cancer care by analyzing vast amounts of data to tailor treatment plans to individual patients [6, 12, 15, 21]. AI algorithms can integrate data from various sources, including genetic information, medical history, and treatment responses, to predict the most effective treatment strategies. For example, AI-driven decision support systems have been developed to recommend personalized treatment regimens for patients with complex cancers, such as metastatic breast cancer or glioblastoma [13, 14, 22, 23]. In radiotherapy planning, AI has shown significant promise by optimizing radiation dose distribution and minimizing exposure to surrounding healthy tissues [15, 24].

Beyond clinical applications, AI plays a pivotal role in cancer research, particularly in target identification and drug discovery [3, 5, 6]. AI-driven algorithms can analyze large datasets of molecular and genetic information to identify new therapeutic targets, accelerating the drug development process. In drug discovery, AI tools can predict how

potential drugs will interact with cancer cells, facilitating the rapid identification of promising compounds for clinical trials [25]. These AI applications significantly reduce the time and cost associated with traditional drug discovery methods, fostering innovation in cancer treatment while potentially increasing the availability of targeted therapies for diverse patient populations.

Summary of Studies on AI-Driven Treatment Algorithms and Their Impact on Different Demographic Groups

While the potential of AI to personalize cancer treatment is substantial, its impact on different demographic groups remains uneven [10, 12, 15, 19, 21]. Studies have shown that AI-driven treatment recommendations can be influenced by the biases present in the training data [9, 12, 15, 16, 18, 21]. For example, AI algorithms developed using epidemiological data, clinical guidelines, and trained with datasets primarily sourced from patients in high-income countries may not perform effectively for patients in low- and middle-income countries. These algorithms often fail to account for the diverse genetic, environmental, social determinants of health data, and unique epidemiological patterns prevalent in these regions, potentially leading to suboptimal diagnosis and treatment recommendations [16, 23, 26–28].

Consideration of Whether AI-Driven Personalized Medicine Is Accessible to All or Limited to Specific Populations

Access to AI-driven personalized medicine is another critical issue. The cost and complexity of implementing AI tools in clinical practice can limit their availability to well-funded healthcare institutions, often leaving underserved populations without access to these advancements [17, 26]. Furthermore, the digital divide—disparities in access to digital technologies—further exacerbates these inequities. In regions lacking robust internet infrastructure or among populations with limited digital literacy, the benefits of AI-driven healthcare are often out of reach [26–28]. Ensuring equitable access to personalized medicine will require addressing these barriers through policies that incentivizes AI deployment in underserved areas, along with targeted funding initiatives, and the development of AI tools that are adaptable to various healthcare settings [27, 28].

AI in Cancer Risk Prediction, Predictive Analytics and Outcomes

AI for Cancer Risk Prediction and Triage

AI has demonstrated considerable potential in cancer risk prediction and triage, enabling earlier identification of individuals at higher risk and optimizing resource allocation in clinical settings. Machine learning models, for instance, can analyze patient data, including genetic markers, lifestyle factors, and family history, to predict an individual's likelihood of developing cancer [6, 10, 23]. These algorithms have been particularly useful in identifying patients at risk for cancers like breast, colorectal, and lung cancer [10, 12, 13, 29, 30]. By prioritizing high-risk patients for screening or further diagnostic procedures, AI aids in triaging care efficiently, potentially improving early detection rates and outcomes. However, the performance of AI in these tasks is highly dependent on the quality and diversity of the datasets, with biases in risk prediction models posing a significant concern for underserved populations.

Evaluation of AI Tools Used for Predicting Cancer Prognosis and Treatment Outcomes

Predictive analytics is one of the most promising areas of AI application in cancer care, offering the potential to predict patient outcomes, such as survival rates, treatment responses, and the likelihood of recurrence [13, 14, 18, 22, 23, 27]. AI models, particularly those using deep learning, have been developed to analyze complex multimodal datasets—including genomic data, clinical notes, laboratory data, social determinants of health data and imaging data—to make accurate predictions about cancer outcomes [13, 18, 22, 27]. These tools can help clinicians identify high-risk patients and tailor interventions to improve outcomes.

Prediction Accuracy Across Different Racial, Ethnic, and Socio-Economic Groups

Research indicates that AI models may perform differently based on the racial, ethnic, gender and socio-economic characteristics of the patients [9, 11, 18, 20, 27]. For instance, studies have found that AI models trained on predominantly White populations tend to be less accurate when applied to Black or Hispanic patients in the U.S. [15–19]. This discrepancy can lead to disparities in cancer prognosis, where certain groups may receive less accurate predictions, potentially influencing treatment decisions and outcomes.

Potential of AI to Perpetuate Existing Biases in Predicting Cancer Outcomes

The potential for AI to perpetuate or even exacerbate existing biases in cancer care is a significant concern. If not carefully designed and validated, AI models may reinforce existing disparities by replicating the biases present in historical data. For example, if an AI model learns from a dataset where certain racial or socio-economic groups have worse outcomes due to historical inequities, it may predict worse outcomes for similar patients in the

future, thereby influencing clinical decisions in a way that perpetuates those disparities [31, 32]. Addressing these biases requires careful consideration in model development, fine-tuning and continuous periodic validation, including the use of diverse and representative datasets and the implementation of bias mitigation strategies, such as model auditing, output explainability models, and the use of carefully generated synthetic data [31, 33]. It is essential to note that the quality of synthetic data is crucial- as poorly generated synthetic data can introduce new biases or fail to adequately represent the diversity needed.

Barriers to Equitable AI Implementation

Data Quality, Representativeness, and Access to Technology

The equitable implementation of AI in cancer care is hindered by several barriers, including data quality, representativeness the data used to train AI models, and access to technology [31] (Table 1). Many AI systems rely on data from electronic health records, clinical trials, or medical imaging repositories that may not fully represent the diversity of the general population [31]. For example, clinical trials often underrepresent Black and Hispanic groups, which leads to limited efficacy and safety data for these populations [34]. Consequently, AI models trained on such data may not provide accurate or personalized recommendations for underrepresented groups, even when bias mitigation strategies are in place. This lack of representativeness can lead to AI models that are less effective for certain groups, particularly those that are underrepresented in the training data. Moreover, issues related to data quality, such as incomplete or inconsistent records, can further compromise the performance of AI tools [31].

Review of Studies on Biases in AI Models and Their Impact on Cancer Care for Under-represented Populations

The biases inherent in AI models can have profound impacts on cancer care for underserved populations. Several studies have documented cases where AI algorithms have performed worse for certain racial or ethnic groups, leading to disparities in diagnosis, treatment, and outcomes. Algorithms that were designed for conditions like kidney function have shown racial bias by overestimating the kidney health of Black patients leading to delays in their treatment and referral for transplants [15]. Another study found that an AI tool used to allocate healthcare resources was less likely to identify Black patients as needing additional care, despite their higher burden of chronic illness [17]. These findings highlight the importance of addressing biases in AI models to prevent the exacerbation of health disparities.

Examination of Structural, Systemic, and Policy-Level Barriers Affecting the Equitable Use of AI in Cancer Care

Beyond the technical challenges, there are structural, systemic, and policy-level barriers that affect the equitable use of AI in cancer care. These include the lack of regulatory frameworks to ensure the fairness and transparency of AI tools, as well as the need for policies that promote the inclusion of diverse populations in AI research [31, 32]. Current regulations like the Health Insurance Portability and Accountability Act (HIPAA) address data privacy but fall short of covering AI's unique risks, such as algorithmic bias [31, 32, 35]. Additionally, systemic issues such as unequal access to healthcare, digital literacy, and the availability of resources to implement AI technologies in underserved areas must be addressed to ensure that AI benefits all patients.

Table 1 Summary of key challenges and opportunities for artificial intelligence (AI) in addressing cancer health equity

Category	Challenges	Opportunities
Data Bias	- Underrepresentation of medically underserved and socially disadvantaged patient communities in datasets - Biases in AI models replicating systemic inequities	- Development of inclusive datasets - Use of carefully designed and validated synthetic data, that mimics diverse real-world conditions, making models more robust and to fill gaps
Access to Technology	- Limited access in low-income or rural areas - Digital divide in access to AI tools	- Policies promoting AI access in underserved areas - Financial support for implementation
AI Model Performance	- Reduced accuracy for underrepresented groups - Generalizability across diverse populations	- Improved model validation - Incorporation of social determinants of health
Policy and Regulation	- Lack of robust regulatory frameworks for bias mitigation - Limited transparency in AI decision-making	- Development of ethical guidelines for transparency - Policies to support diverse research
Ethical Considerations	- Risk of exacerbating disparities in resource allocation - Misuse of biased predictions	- Ethical frameworks ensuring fairness - Educating healthcare providers and health system leaders on responsible AI use

Programs like the Affordable Connectivity Program (ACP) should be renewed to close the digital divide and ensure that underserved communities benefit from AI advancements [36]. Policymakers, healthcare providers, and AI developers must collaborate to update regulations, promote inclusive research, and enhance digital access and skills for all patients.

Discussion

Assessment of Current Evidence

The current literature reveals both the potential and challenges of applying AI in cancer care, particularly regarding health equity. AI has demonstrated success in improving cancer diagnosis, treatment planning, and predictive analytics, offering the potential to enhance precision medicine [37, 40–47]. However, evidence also shows that AI models often perform unevenly or unfairly across different demographic groups, primarily due to biased training datasets.

While AI is often seen as a tool for personalized medicine, its benefits are not equally accessible to all. High costs limited technological infrastructure, and the digital divide restrict AI-driven innovations to resource-rich healthcare settings, further widening the gap between advantaged and underserved populations.

AI and the Risk of Widening Disparities

The risk of AI widening health disparities is significant, particularly because of the biases inherent in the data used to train these models. Historical inequities in healthcare data are often reflected in AI algorithms, which may replicate or even amplify these biases. For instance, AI tools used to predict cancer prognosis or treatment outcomes may deliver less accurate results for underrepresented groups, leading to suboptimal care [18, 20, 31].

Additionally, the deployment of AI technologies is often concentrated in high-income, urban areas, leaving rural or low-income populations with limited access to these innovations. As a result, the benefits of AI are disproportionately distributed, with underserved populations falling further behind [17, 25, 29, 37–39].

Potential for AI to Bridge the Divide

Despite these concerns, AI holds promise for bridging the divide in cancer health equity when developed and implemented with a focus on inclusivity, fairness, and accessibility (Table 1). There are notable examples where AI is being used ethically and successfully to improve equity in cancer care. For instance, there are promising AI-driven screening and diagnostic tools designed to address gaps in care for diverse and underrepresented populations [13, 45–47] (Table 2).

AI has the potential to address the social determinants of health by identifying high-risk patients based on factors such as socioeconomic status, access to care, and environmental exposures. When AI models are trained on representative datasets and integrated with data on social determinants of health, they can help reduce disparities in cancer outcomes by enabling more targeted interventions [29, 30, 45–47].

Ethical and Policy Considerations

The ethical implications of AI in cancer care are significant, particularly when it comes to health equity. Transparent and explainable AI algorithms are critical to ensure that healthcare providers understand the decisions these tools make and can identify potential biases [31, 32, 38, 39]. Policies must promote the inclusion of diverse populations in AI research and development, ensuring that the benefits of AI extend to all groups. Furthermore, regulatory frameworks are needed to guarantee that AI tools undergo rigorous testing for bias and are continually monitored for their performance across various demographic groups.

To overcome the digital divide and ensure equitable access to AI-driven cancer care, targeted investments in infrastructure and digital literacy are essential. Policymakers, healthcare providers, and AI developers must collaborate to ensure that AI tools are accessible and effective in both well-resourced and underserved healthcare settings.

Finally, there is a need for education and training for healthcare providers on the ethical use of AI in cancer care. Clinicians must be aware of the potential biases in AI tools and be equipped to use these technologies in ways that promote equity.

Research Gaps

The literature on AI in cancer care reveals several critical gaps that must be addressed to ensure these technologies contribute to health equity. One of the most pressing issues is the lack of diverse and representative datasets used to train AI models. Most AI tools are developed using data from predominantly White, affluent populations, limiting the generalizability of these models to more diverse groups. This lack of representativeness can lead to biases in AI predictions and interventions, worsening existing health disparities.

Another significant gap is the limited research on AI's long-term impact on health equity and financial toxicity in cancer care. Most studies focus on immediate performance measures, like diagnostic accuracy or treatment

Table 2 Selected list of organizations engaged in promising practices leveraging AI tools to promote cancer health equity

Organization	Activity	Website or Reference
Aidence (Netherlands)	Netherlands	Aidence provides AI tools for lung cancer detection via medical imaging. Their mission includes making these tools accessible to a broad range of healthcare systems, ensuring that diverse populations benefit from early detection https://www.aidence.com/
CancerIQ	United States	CancerIQ offers a precision prevention platform that enables healthcare providers to proactively engage patients, assess cancer risk, and implement the latest evidence-based strategies for early intervention. The platform includes a comprehensive library of evidence-based guidelines and risk models, helping providers interpret patient data and guide individuals toward the appropriate genetic testing, screenings, or treatments for cancer prevention and care https://www.canceriq.com/
Ceará Cancer Institute Instituto do Câncer do Ceará, (ICC)	Brazil	The Ceará Cancer Center leverages technologies, including artificial intelligence, to improve cancer diagnostics, treatment planning, and patient outcomes. The center serves as a hub for oncology care in the region, offering state-of-the-art treatments and therapies, while also focusing on increasing its longstanding access to cancer care for underserved populations in Brazil The Korina tool promotes AI-enabled cancer patient population navigation and improved patient outcomes by providing case-specific medical evidence synthesis to care providers https://icc.org.br/ https://www.scielo.br/j/hcsm/a/4MxLjM9LxBTgvyqg59CLK3c/?lang=en
COTA, Inc	United States	COTA leverages their COTA real-world database to explore differences in tumor biology, treatment patterns, health outcomes to gain insights into cancer health disparities affecting underserved populations https://cotahealthcare.com/ https://cotahealthcare.com/real-world-data-and-its-role-in-addressing-cancer-health-disparities/ https://www.hcinnovationgroup.com/population-health-management/health-equity/article/21270788/qa-with-cotas-ck-wang-on-addressing-disparities-in-oncology
CureMetrix	United States	CureMetrix is developing AI-based mammography tools to improve early breast cancer detection. The company is focused on reducing false negatives and improving diagnostic outcomes in diverse populations https://curemetrix.com/our-company-2/
DataPathology	Morocco, Democratic Republic of Congo, Senegal	DataPathology is a healthtech company focused on revolutionizing pathology services through the use of digital tools and artificial intelligence. Their platform streamlines cancer diagnostics by providing remote access to digital pathology images, enabling faster and more accurate diagnoses. DataPathology aims to enhance global access to pathology services, particularly in regions with limited access to specialist care, ensuring timely and equitable cancer diagnosis for all patients https://www.datapathology.ma/

Table 2 (continued)

Organization	Activity	Website or Reference
Flatiron	United States	Flatiron specializes in using real-world data and artificial intelligence to accelerate cancer research and improve patient outcomes. By aggregating and analyzing clinical data from cancer patients, Flatiron helps oncologists and researchers gain insights into treatment effectiveness and patient responses. The company focuses on making cancer care more personalized and accessible by integrating AI into clinical workflows and supporting data-driven decision-making in oncology https://flatiron.com/
Foundation for Innovative new Diagnostics (FIND)	United States	FIND uses AI to develop innovative cancer diagnostics, with funding from the Gates Foundation. Their work focuses on expanding access to cancer screening in underserved regions https://www.finddx.org/?s=cancer
Freenome	United States	Freenome uses AI to detect early-stage cancer through non-invasive blood tests. Their approach integrates AI with multi-omic data (genomic, proteomic, etc.) to develop screening tests that are equally effective across diverse populations https://www.freenome.com/
Hologic	United States	Hologic is a global medical technology company specializing in women's health, with a strong focus on early detection and treatment of diseases such as breast cancer. Hologic develops and provides advanced diagnostic tools, including AI-powered imaging solutions, to enhance accuracy in breast cancer screening and improve patient outcomes. The company is committed to reducing healthcare disparities by making innovative diagnostic technologies more accessible worldwide https://www.hologic.com/project-health-equity
Hurone AI	United States, Rwanda	The company's AI platform provides oncology care solutions tailored for community oncology centers in the United States and adaptable worldwide, including in low- and middle-income countries, addressing cancer treatment challenges in resource-constrained environments. Hurone AI's system is designed to assist oncologists with treatment plans, predict patient outcomes, and optimize follow-up care, especially in regions where access to oncology specialists is limited https://hurone.ai/
Ibex Medical Analytics	Israel	Ibex uses AI to assist in cancer diagnosis, particularly in pathology. Their AI solutions are designed to improve the accuracy and speed of cancer detection across diverse populations. AI-based cancer diagnostic tools that have been deployed in various global markets, improving diagnostic precision in underserved regions https://ibex-ai.com/about/
Institute for Health Metrics and Evaluation (IHME)	United States	IHME uses AI and big data to map cancer incidence and outcomes globally, with a focus on low- and middle-income countries. Notable projects include the use of AI models to assess the burden of cancer in underserved populations https://www.healthdata.org/

Table 2 (continued)

Organization	Activity	Website or Reference
Intixel	Egypt	Intixel is a healthcare technology company specializing in AI-powered medical imaging solutions, with a focus on enhancing cancer detection and diagnosis. Their platform utilizes artificial intelligence to assist radiologists in interpreting medical images with greater accuracy and efficiency. Intixel's solutions aim to improve early cancer detection rates, reduce diagnostic errors, and streamline the imaging workflow, particularly in areas with high patient volumes and limited access to expert radiologists https://www.intixel.com/
Kheiron Medical	United Kingdom	Kheiron Medical develops AI-powered breast cancer screening software that aims to make mammography more accessible and reliable, especially in low-resource settings https://www.kheironmed.com/
Lunit	South Korea	Lunit develops AI software for cancer detection and treatment through medical imaging. Their focus includes making these technologies accessible in regions with fewer healthcare resources, ensuring that AI can help reduce global disparities in cancer care https://www.lunit.io/en/company
MinoHealth AI Labs	Ghana	MinoHealth focuses on using AI to improve healthcare in Africa, including cancer diagnosis. Their AI-driven healthcare platform helps clinicians in early cancer detection and diagnostics, aimed at reducing disparities in Africa https://www.minohealth.ai/ https://www.minohealth.ai/#about
MultiplAI Health	United Kingdom	MultiplAI Health is transforming cancer care and other complex disease detection by combining cutting-edge genomics with AI. Their approach involves a universal blood test that uses RNA sequencing and AI to screen for a wide range of diseases, including cardiovascular conditions and cancer. This whole blood test offers a non-invasive, remote screening option, allowing for earlier detection and more personalized care. By leveraging AI, MultiplAI Health aims to simplify and enhance the accuracy of diagnostics, making it easier for healthcare providers to identify diseases before they progress https://www.multiplaihealth.com/#Home
Neuralabs Africa	Senegal, Kenya	Neuralabs Africa is a tech-driven company focused on harnessing artificial intelligence to improve healthcare across the continent, with an emphasis on cancer care and diagnostics. Their AI-powered platforms aim to bridge the gap in access to high-quality diagnostic tools by providing advanced medical imaging solutions to underserved regions. Neuralabs Africa's goal is to improve early detection and accurate diagnosis of cancer, leading to better treatment outcomes and addressing healthcare disparities in Africa https://neurallabs.africa/
Oncora Medical (USA)	United States	Oncora Medical AI-enhanced decision-support system is designed to improve treatment accuracy across various demographic groups, mitigating disparities in radiotherapy outcomes. Oncora Medical develops AI-powered software to improve radiation oncology treatment plans and focus on collecting diverse data sets to ensure their algorithms serve minority populations equitably https://www.oncora.ai/ https://www.oncora.ai/research

Table 2 (continued)

Organization	Activity	Website or Reference
Owkin	France	Owkin uses AI to analyze medical images and genetic data to improve cancer diagnosis and treatment. The company collaborates with hospitals to ensure diverse datasets are used to reduce bias in their algorithms https://www.owkin.com/
Paige AI	United States	Paige AI develops AI software that assists pathologists in detecting cancer. They focus on improving access to high-quality diagnostics, especially in underserved regions, by reducing the reliance on highly specialized pathologists They develop AI models for cancer diagnosis with a focus on inclusive training datasets that represent various population groups, aiming to improve diagnostic equity in pathology https://paige.ai/
Panakeia AI	United Kingdom	Panakeia simplifies and speeds up cancer diagnosis by reducing the need for multiple tests, making the process more affordable. Their breast cancer diagnostic tool analyzes H&E slides to determine tissue ER, PR, and HER2 status. The tool integrates easily with existing digital workflows, eliminating the need for time-consuming lab tests and providing accurate results more quickly to patients and healthcare teams https://www.panakeia.ai/
PapsAI	Uganda	PapsAI is a healthtech start-up in Uganda focused on leveraging artificial intelligence to enhance cervical cancer screening and diagnostics. By using AI-powered tools, PapsAI aims to improve the accuracy and accessibility of Pap smear tests, especially in low-resource settings. The platform is designed to support healthcare providers in early detection of cervical cancer, ensuring timely intervention and better patient outcomes in underserved communities https://papsai.com/
PathAI	United States	PathAI uses AI to improve diagnostic accuracy. Their work includes cancer diagnostics, aiming to reduce errors and ensure timely treatment in low-resource settings. They have developed AI-based pathology platforms that are designed to reduce variability in cancer diagnostics, particularly for minority groups that are often underrepresented in clinical data https://www.pathai.com/about-us/
Patient Discovery	United States	Patient Discovery is focused on improving patient engagement and health outcomes by leveraging AI and behavioral science. The company's platform empowers patients by collecting real-time data on their health experiences, social determinants, and treatment barriers, providing actionable insights for healthcare providers. By integrating patient-reported outcomes with AI-driven analytics, Patient Discovery helps ensure more personalized care and enables healthcare teams to address inequities, particularly for patients with cancer https://www.patientdiscovery.com/ https://www.patientdiscovery.com/resources/patient-discovery-partners-with-amerisourcebergen-for-cancer-care-equity

Table 2 (continued)

Organization	Activity	Website or Reference
Rology	Egypt, Kenya, South Africa	Rology is a teleradiology platform that leverages artificial intelligence to connect radiologists with healthcare providers, particularly in regions with a shortage of specialists. By using AI to streamline the radiology workflow, Rology helps deliver faster and more accurate medical imaging diagnoses, including for cancer detection. The platform aims to improve access to high-quality radiology services in underserved areas, ensuring timely diagnoses and better patient outcomes https://rology.health/
Savor Health	United States	Savor Health is a B2B digital therapeutic platform that provides personalized nutritional strategies to help cancer patients manage their symptoms and side effects. Their AI-driven tool, *Ina*®, offers tailored nutrition and symptom management guidance, empowering patients to take control of their health. *Ina*® also supports caregivers, known as the *Casserole Patrol*™, by offering recipes, menu plans, and tips to help them provide the most appropriate meals and care for their loved ones with cancer https://savorhealth.com/what-we-do/
SkinVision	Netherlands	SkinVision uses AI for early detection of skin cancer through a mobile app that allows users to scan moles and skin lesions. The platform is designed to be accessible to people globally, regardless of geographic or socioeconomic status Their goal is to expand access to quality diagnostics in regions where specialized dermatologists are scarce https://www.skinvision.com/
Tempus AI	United States	AI-driven precision oncology platform to deliver personalized cancer care, focusing on including underrepresented populations in their dataset for more equitable treatment outcomes https://www.tempus.com/resources/content/articles/qa-using-ai-to-identify-and-close-gaps-in-care-for-cancer-patients/?srsltid=AfmBOorajBr8HI90p6skVDVzKQ8IIRv7EphMIsmS06Zy9yTqQX4yDb6t https://www.tempus.com/oncology/care-pathway-solutions/?srsltid=AfmBOorMhJIQ1ZZgxlRJYyGAD5ur5m66ZQmKse7XvYilXyfOVHqZ-uZI
Vectorgram Health	Kenya	Vectorgram Health is focused on enhancing cancer diagnostics and care through artificial intelligence and cloud-based solutions. Their AI-powered platform aims to assist healthcare providers in detecting cancer earlier and more accurately by analyzing medical images and pathology reports. The company is dedicated to improving access to advanced diagnostic tools, particularly in sub-Saharan Africa, where there are often shortages of specialists and diagnostic infrastructure https://www.vectorgram.co/

recommendations, without considering broader, systemic effects. There is a need for longitudinal studies that examine how AI influences outcomes and financial burdens over time, particularly in underserved populations.

Additionally, research integrating social determinants of health into AI models is scarce. Although some studies explore how socioeconomic status, access to care, and environmental factors influence cancer outcomes, these variables are often not incorporated into AI algorithms. This omission limits AI's ability to fully address the root causes of health disparities.

Future Research Directions

To address these gaps, future research must prioritize the development of AI models that are inclusive and representative of diverse populations. This requires concerted efforts to collect and curate datasets that reflect the racial,

ethnic, socioeconomic, and geographic diversity of the broader population. Collaboration between researchers, healthcare providers, and communities is essential to ensure that AI tools are developed with a deep understanding of the unique challenges faced by different groups. Implementing data-sharing agreements that comply with privacy standards and data regulations, such as HIPAA and the General Data Protection Regulation (GDPR), is essential. These agreements should facilitate access to diverse datasets while maintaining patient confidentiality. Additionally, the use of carefully designed and validated synthetic data—i.e., data generated to mimic real patient data without revealing personal information—can be employed to supplement real-world datasets and overcome privacy concerns.

Moreover, future studies should focus on the ethical design of AI in cancer care. This includes not only addressing biases in datasets but also developing algorithms that are transparent and explainable, so that healthcare providers can understand and trust AI-driven decisions. Researchers should explore techniques for mitigating biases, such as algorithmic fairness and bias correction methods, and rigorously test these approaches in diverse populations.

Interdisciplinary research that combines AI with social determinants of health is another crucial area for future exploration. By integrating data on social, economic, and environmental factors into AI models, researchers can develop more comprehensive tools that account for the broader context of patients' lives. Such models have the potential to improve predictions, tailor interventions more effectively, and ultimately reduce disparities in cancer outcomes.

In addition, future research should also examine the policy and regulatory frameworks needed to support equitable AI implementation in cancer care. This includes evaluating the effectiveness of existing guidelines and proposing new policies to ensure that AI tools are used responsibly and fairly. Researchers should advocate for regulations that encourage data-sharing agreements, ensure compliance with privacy laws, and promote the use of appropriate data to include diverse populations in AI research without compromising patient privacy.

Finally, there is a need for research on strategies to overcome the digital divide and ensure that AI-driven innovations are accessible to all patients, regardless of their socioeconomic status or geographic location. This could include studies on the implementation of AI tools in resource-limited settings, as well as interventions to improve digital literacy among patients and healthcare providers.

Policy Implications

To ensure that AI contributes to health equity in cancer care, policymakers must take proactive steps to address the challenges identified in the current literature. This includes creating incentives for the development of inclusive AI models, such as funding initiatives that support the collection of diverse datasets and the development of bias-mitigation techniques [31].

Policies should also focus on expanding access to AI technologies in underserved areas. This could involve providing financial support for healthcare institutions in low-income or rural regions to adopt AI tools, as well as investing in the necessary infrastructure to support these technologies (Table 1).

Moreover, regulatory bodies should establish standards for the ethical use of AI in healthcare, including guidelines for transparency, accountability, and fairness. These standards should require that AI tools undergo rigorous testing for biases and that their performance is regularly monitored to ensure they do not contribute to health disparities (Table 1).

Educational programs for healthcare providers on the use of AI in clinical practice are also essential. By equipping clinicians with the knowledge and skills to use AI responsibly, policymakers can help ensure that these technologies are used in ways that promote, rather than hinder, health equity.

Conclusion

AI represents a transformative force in cancer care, offering unprecedented opportunities to enhance diagnosis, treatment, and predictive analytics. However, the potential of AI to bridge the existing disparities in cancer outcomes is accompanied by significant challenges that must be addressed to ensure equity.

Our review highlights the dual-edged nature of AI in cancer care: while AI-driven technologies can improve precision and personalized treatment, they also risk perpetuating or exacerbating health disparities. The biases inherent in training datasets, the underrepresentation of diverse populations in AI research, and the unequal access to AI technologies all contribute to the uneven benefits of these advancements. Encouraging the establishment of ethical data-sharing agreements that respect privacy laws, utilize synthetic data, and foster collaboration among research institutions, healthcare providers, and community organizations can help create more comprehensive and inclusive datasets.

To truly harness AI as a tool for promoting cancer health equity, the healthcare and research communities must take

deliberate actions. This includes developing and using inclusive, representative datasets; integrating social determinants of health into AI models; and establishing ethical guidelines that ensure AI tools are transparent, fair, and accessible to all. Moreover, policies must be implemented to overcome the digital divide and ensure that AI-driven innovations reach underserved populations.

The future of AI in cancer care hinges on our ability to address these challenges. By prioritizing equity in the development and deployment of AI technologies, we can work towards a healthcare system where all patients, regardless of their background, can benefit from the advancements in AI. The findings of this review underscore the need for continued research, interdisciplinary collaboration, and policy interventions that are focused on reducing disparities and promoting health equity. AI has the potential to be a powerful force for good in cancer care, but realizing this potential requires a commitment to ensuring that no population is left behind.

Key References

Annotated Reference List of Selected (Recently Published) Studies

Overview of Studies Assessing AI in Cancer Risk Prediction and Triage

- Yala A, Mikhael PG, Strand F, Lin G, Satuluru S, et al. Multi-Institutional Validation of a Mammography-Based Breast Cancer Risk Model. J Clin Oncol. 2021 Jun 1;40(16):1732–1740. https://doi.org/10.1200/JCO.21.01337.
  - The paper presents a study validating the AI-based breast cancer risk model called *Mirai*. The purpose of the study is to assess a deep learning model's (Mirai) effectiveness in predicting breast cancer risk across diverse global populations, including screening populations from hospitals in the United States, Israel, Sweden, Taiwan, and Brazil.
- Yala A, Mikhael P, Strand F, Lin G, Smith K, Wan Y-L, Lamb L, Hughes K, Lehman C, Barzilay R, Toward robust mammography-based models for breast cancer risk. *Sci. Transl. Med.*13, eaba4373(2021). https://doi.org/10.1126/scitranslmed.aba4373.
  - The researchers developed Mirai, a mammography-based deep learning model designed to predict 5-year breast cancer risk and high-risk patients at multiple timepoints and across three countries – US, Sweden, and Taiwan, and across race and ethnicity categories. This model improves upon conventional risk factors like family history, genetic predisposition, or age, and focuses on creating personalized risk profiles based solely on imaging data. By training on large, diverse datasets, the model shows promise in identifying high-risk patients who might benefit from more frequent screenings or preventive interventions.
- Yu C, Helwig EJ. The role of AI technology in prediction, diagnosis and treatment of colorectal cancer. *Artif Intell Rev*. 2022;55(1):323–343. https://doi.org/10.1007/s10462-021-10034.
  - The paper provides a detailed exploration of how artificial intelligence (AI) technologies, including machine learning (ML) and deep learning (DL), are revolutionizing colorectal cancer (CRC) care. Key areas of focus include epidemiological prediction, diagnosis through imaging technologies like endoscopy and MRI, genetic diagnosis, and AI-driven treatment recommendations. The paper also highlights the potential for AI to democratize cancer care by providing more precise diagnostic and treatment tools, especially in geographically and economically underserved regions. Technologies like GeoAI can help target interventions where CRC incidence is highest, and AI-powered diagnostic tools can improve access to high-quality care, regardless of location.
- Kumar Y, Gupta S, Singla R. Hu Y-C: A systematic review of artificial intelligence techniques in cancer prediction and diagnosis. *Arch Comput Methods Eng*. 2021;2021:1–28.
  - This systematic review presents an in-depth review of the use of AI technologies, including machine learning (ML) and deep learning (DL), in cancer care. It examines how AI models contribute to cancer prediction, diagnosis, risk assessment, and survivability prediction. AI has shown significant promise in enhancing the accuracy, speed, and reliability of cancer diagnoses by processing complex data, such as imaging and genomic information.

Overview of Selected Studies on Recent Advances in AI for Cancer Care and Diagnosis

- Wu Q, Huang Y, Wang S, Qi L, Zhang Z, Hou D, et al. Artificial intelligence in lung cancer screening: Detection, classification, prediction, and prognosis. Cancer Med. 2024 Apr;13(7). https://doi.org/10.1002/cam4.7140.
  - This paper reviews the role of AI, especially convolutional neural networks (CNNs), in improving lung cancer screening and diagnosis. AI enhances lung seg-

mentation, nodule detection, false-positive reduction, nodule classification, and predicting cancer prognosis. Despite these advancements, challenges such as limited datasets and the interpretability of results remain. This is critical for AI in health equity as it improves early cancer detection, especially for underserved populations, but further validation and inclusivity in data are required to prevent biases in care delivery.

- de Margerie-Mellon C, Chassagnon G. Artificial intelligence: A critical review of applications for lung nodule and lung cancer. Diagn Interv Imaging. 2023 Jan;104(1):11–17. https://doi.org/10.1016/j.diii.2022.11.007. Epub 2022 Dec 10. PMID: 36,513,593.
  - The paper provides a comprehensive overview of AI's role in medical imaging for lung cancer, particularly focusing on lung nodule detection, segmentation, and classification using deep learning techniques. AI applications have demonstrated improved accuracy in detecting and classifying lung nodules from imaging scans, aiding early cancer diagnosis. However, challenges remain in clinical implementation due to the lack of generalizability, "black box" algorithms, and the need for further validation.
- Ong W, Lee A, Tan WC, Fong KTD, Lai DD, Tan YL, Low XZ, Ge S, Makmur A, Ong SJ, et al. Oncologic Applications of Artificial Intelligence and Deep Learning Methods in CT Spine Imaging—A Systematic Review. *Cancers*. 2024; 16(17):2988. https://doi.org/10.3390/cancers16172988
  - This systematic review highlights the potential of AI and deep learning in spinal oncology, with a particular focus on CT imaging. AI applications improve the accuracy of cancer detection, including prostate cancer, by identifying malignant cells more effectively. These methods enhance early diagnosis and help guide treatment planning.
- Seyrek FB, Yiğit H, Diagnosis of Lung Cancer from Computed Tomography Scans with Deep Learning Methods. *Journal of Universal Computer Science*, 2024: 1089–1111.
  - The study shows that deep learning models applied to CT scans significantly improve the detection of lung cancer. These AI models outperform traditional diagnostic methods by offering more precise detection of abnormalities in lung tissues, contributing to earlier and more reliable lung cancer diagnosis.
- Wang L. Mammography with deep learning for breast cancer detection. Front Oncol. 2024 Feb 12;14:1,281,922. https://doi.org/10.3389/fonc.2024.1281922. PMID: 38,410,114; PMCID: PMC10894909.
  - This study reviews how deep learning enhances mammography's ability to detect breast cancer by improving accuracy, reducing false positives, and enabling more personalized patient care. It discusses various deep learning architectures like CNNs and their applications in breast lesion detection and classification. The study offers the potential to make breast cancer screening more accessible and reliable, especially for underrepresented groups, improving early detection and potentially reducing disparities in outcomes.
- Hernández-Vázquez MA, Hernández-Rodríguez YM, Cortes-Rojas FD, Bayareh-Mancilla R, Cigarroa-Mayorga OE. Hybrid Feature Mammogram Analysis: Detecting and Localizing Microcalcifications Combining Gabor, Prewitt, GLCM Features, and Top Hat Filtering Enhanced with CNN Architecture. *Diagnostics*. 2024; 14(15):1691. https://doi.org/10.3390/diagnostics14151691.
  - This study combined traditional feature extraction techniques with CNNs to detect and localize microcalcifications in mammograms, an early indicator of breast cancer. The integration of these methods significantly improved the precision and reliability of automated mammographic analysis, enhancing the support provided to radiologists in early breast cancer detection. The hybrid CNN model increased accuracy in detecting subtle signs of cancer, improving diagnostic outcomes and supporting radiologists in identifying early-stage breast cancer.
- Razali NF, Isa IS, Sulaiman SN, Osman MK, Karim NKA and Nordin, SA "Optimization of ReLU Activation Function for Deep-Learning-based Breast Cancer Classification on Mammogram Images," *2024 IEEE International Conference on Automatic Control and Intelligent Systems (I2CACIS)*, Shah Alam, Malaysia, 2024, pp. 267–272, https://doi.org/10.1109/I2CACIS61270.2024.10649623.
  - In this study, the researchers improved the ReLU activation function in convolutional neural networks (CNNs) to enhance the accuracy of deep learning models used for breast cancer classification in mammogram images. By optimizing the ReLU function, the model's accuracy in detecting breast cancer increased from 95.0% to 96.5%. This improvement shows that the optimized model performs as well as, or even better than, radiologists in identifying breast cancer through mammograms.

  ReLU, or Rectified Linear Unit, is an activation function commonly used in deep learning, particularly in

convolutional neural networks (CNNs). It's a simple yet highly effective function that helps a neural network learn complex patterns in data.

Studies Assessing AI in Cancer Research and Accelerated Drug Discovery

- Zadeh Shirazi A, Tofighi M, Gharavi A, Gomez GA. The Application of Artificial Intelligence to Cancer Research: A Comprehensive Guide. *Technology in Cancer Research & Treatment*. 2024;23. https://doi.org/10.1177/15330338241250324.
  - This paper highlights how AI is transforming cancer research through advancements in machine learning, deep learning, and soft computing. It explores AI's contributions to cancer diagnosis, treatment prediction, and patient monitoring by improving detection accuracy, reducing diagnostic errors, and facilitating personalized care. The relevance of AI to health equity lies in its potential to provide consistent and accurate cancer care across diverse populations. However, inclusivity in data and algorithm design remains crucial to avoid biases that could perpetuate healthcare disparities.
- Paul D, Sanap G, Shenoy S, Kalyane D, Kalia K, Tekade RK. Artificial intelligence in drug discovery and development. Drug Discov Today. 2021 Jan;26(1):80–93. https://doi.org/10.1016/j.drudis.2020.10.010.
  - This review explores the growing impact of AI in the pharmaceutical industry. It highlights AI's applications in drug discovery, including drug repurposing, improving pharmaceutical productivity, and streamlining clinical trials. By automating and accelerating complex processes, AI reduces the human workload and shortens drug development timelines.
    In terms of health equity, AI can democratize access to innovative treatments by optimizing resource use and reducing costs. However, ensuring that AI models are trained on diverse datasets is crucial to avoid bias and inequities in drug development and distribution.

Studies Assessing the Accuracy and Efficiency of AI Tools in Diverse Populations

- Chen RJ, Wang JJ, Williamson DFK, Chen TY, Lipkova J, Lu MY, Sahai S, Mahmood F. Algorithmic fairness in artificial intelligence for medicine and healthcare. Nat Biomed Eng. 2023 Jun;7(6):719–742. https://doi.org/10.1038/s41551-023-01056-8. Epub 2023 Jun 28. PMID: 37,380,750; PMCID: PMC10632090.
  - In this perspective, the authors outlined how the data generation, representation, model building, training and performance of AI algorithms, can lead to inequities and health disparities, without standards for fair AI in medicine and healthcare.
- Yala A, Mikhael P, Strand F, Lin G, Smith K, Wan Y-L, Lamb L, Hughes K, Lehman C, Barzilay R, Toward robust mammography-based models for breast cancer risk. *Sci. Transl. Med.*13, eaba4373(2021).https://doi.org/10.1126/scitranslmed.aba4373
  - The researchers developed Mirai, a mammography-based deep learning model designed to predict 5-year breast cancer risk and high-risk patients at multiple timepoints and across three countries – US, Sweden, and Taiwan, and across race and ethnicity categories. This model improves upon conventional risk factors like family history, genetic predisposition, or age, and focuses on creating personalized risk profiles based solely on imaging data. By training on large, diverse datasets, the model shows promise in identifying high-risk patients who might benefit from more frequent screenings or preventive interventions.
- Omoleye OJ, Woodard AE, Howard FM, Zhao F, Yoshimatsu TF, Zheng Y, Pearson AT, Levental M, Aribisala BS, Kulkarni K, Karczmar GS, Olopade OI, Abe H, Huo D. External Evaluation of a Mammography-based Deep Learning Model for Predicting Breast Cancer in an Ethnically Diverse Population. Radiol Artif Intell. 2023 Jul 26;5(6):e220299. https://doi.org/10.1148/ryai.220299. PMID: 38,074,785; PMCID: PMC10698602.
  - The researchers leveraged Mirai, the mammography deep learning AI model to evaluate its performance in a high-risk racially diverse African American population in the US. The model performed well in breast cancer prediction with good performance for the selected high-risk case control population that included BRCA mutation carriers and African American women.

AI and Health Equity in Resource-Limited Settings

- West SM, Whittaker M, Crawford K. Discriminating systems: gender, race, and power in AI. AI Now Institute. 2019.
  - This foundational report explores the intersections of gender, race, and power dynamics in the development and application of artificial intelligence (AI) systems. The authors highlight systemic biases and the lack of diversity in the AI workforce and argue that these disparities influence the biases embedded in AI technolo-

gies. The document provides a critique of the "pipeline problem" in tech diversity, challenges the fairness of current workplace practices, and emphasizes how biased data perpetuates inequities. The report concludes with specific recommendations for the AI industry, such as increasing workforce transparency, rigorously auditing AI systems for bias, and developing inclusive datasets that represent diverse identities and experiences. This report is a critical resource for understanding the socio-technical implications of AI in exacerbating inequalities. It offers a compelling case for embedding fairness and accountability into AI development and implementation. Researchers, policymakers, and AI developers can use its recommendations to create equitable and inclusive AI practices.

- López DM, Rico-Olarte C, Blobel B and Hullin C. Challenges and solutions for transforming health ecosystems in low and middle-income countries through artificial intelligence. Front Med. 2022;9:958097. https://doi.org/10.1016/j.drudis.2020.10.010.

  - This systematic review examines the role of artificial intelligence (AI) in transforming healthcare systems in low- and middle-income countries (LMICs). Using the PRISMA-ScR guidelines, the authors reviewed 151 studies published between 2011 and 2022. They identify 40 challenges related to AI adoption in LMICs across eight categories: data quality, context awareness, regulation, education and change resistance, financial resources, methodology, infrastructure and connectivity, and scalability.

    Key findings highlight the dominance of machine learning models in healthcare AI research and the frequent use of small datasets from LMICs, which limit model generalizability. The study emphasizes ethical and regulatory gaps, insufficient infrastructure, and resistance to change as major barriers to AI integration in LMIC healthcare. The authors provide 89 actionable recommendations for addressing these challenges, such as improving data quality, adopting context-aware AI models, developing robust ethical and regulatory frameworks, and ensuring sustainable funding for AI projects.

    This article is a critical resource for stakeholders aiming to leverage AI to address global health disparities, offering a comprehensive framework for strengthening health ecosystems in resource-constrained settings.

- Wahl B, Cossy-Gantner A, Germann S, Schwalbe NR. Artificial intelligence (AI) and global health: How can AI contribute to health in resource-poor settings? BMJ Glob Health. 2018;3(4).

  - This article explores the transformative potential of artificial intelligence (AI) in improving healthcare delivery in resource-poor settings. While AI has been extensively applied in high-income countries, its deployment in low- and middle-income countries (LMICs) is relatively nascent. The authors categorize key AI technologies—including machine learning, natural language processing (NLP), expert systems, and automated planning—and examine their potential to address critical healthcare challenges in LMICs, such as inadequate infrastructure, limited access to medical expertise, and resource constraints.

    The article also discusses complementary digital health technologies, including electronic medical records (EMRs), cloud computing, and mobile health (mHealth), which can enhance AI applications. Ethical considerations, such as privacy, informed consent, and the potential for AI bias, are highlighted as critical challenges that must be addressed to ensure equitable AI adoption. The authors propose that scaling AI in LMICs requires locally tailored solutions, robust datasets, and collaborative approaches integrating local expertise.

Author Contribution IDM made substantial contribution to the conception and design of the review. IDM, KN and DA revised the manuscript critically for important intellectual content and added insightful material. All authors reviewed, edited and approved the final manuscript version for submission.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of Interest Irene Dankwa-Mullan provides advisory support for Hurone AI and Patient Discovery; Kingsley Ndoh is the CEO of Hurone AI and has received grant support from Amazon Web Services, Darlington Akogo is the founder and CEO of MinoHealth AI Labs and has received grant support from the Bill and Melinda Gates Foundation, Hermano Alexandre Lima Rocha MD MPH, PhD, and Sergio Juaçaba, MD are in executive senior leadership positions at the Ceará Cancer Institute.

Competing interests All authors are experts and leaders working on the frontlines, actively leveraging AI tools to address disparities in cancer care access within their communities, through their organizations or consultative / advisory work.

Human and Animal Rights and Informed Consent This review paper does not contain any studies with human or animal subjects performed by any of the authors.

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